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Evaluation of Electronic and Paper Textual Glosses on Second Language Vocabulary Learning and Reading Comprehension

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Abstract

Second language studies have supported the use of glossing for enhancing vocabulary learning. Along with technological developments, an increasing number of studies have examined electronic textual glossing in computer-assisted environments, yet only a small number of studies have compared electronic glossing to its traditional paper counterpart. In the present study, we examined the effectiveness of electronic and paper textual glosses on second language vocabulary learning and reading comprehension within a cognitive load theory framework. We adopted an experimental design using Korean undergraduate learners of English as a foreign language, who read a target text under one of the three reading conditions: electronic gloss, paper gloss, and no gloss. The results showed that the electronic glossing condition resulted in the greatest gains of vocabulary. The electronic glossing condition was also the most efficient in light of the cognitive load framework, showing the highest instructional efficiency. However, there were no significant differences among the three reading conditions for measures related to reading comprehension.

Keywords: Electronic gloss · Paper gloss · Reading comprehension · Vocabulary learning

Introduction

Glossing, an instructional method of providing meaning for a selected word in a text, has gained much attention from researchers in the field of second language (L2) studies (e.g., Hulstijn et al. 1996; Jacobs et al. 1994). Nation (2001, 2009) points to the following benefits of enhancing L2 text with glossary information: (1) it enables one to read difficult text that otherwise would be too challenging; (2) it prevents readers from making a wrong guess of vocabulary meaning; and (3) it increases the likelihood of the glossed word being learned. Overall, L2 researchers have identified glossing as a useful and effective form of scaffolding.

In more recent years, researchers have shifted their attention toward the application of textual glosses in the computer-assisted reading environment (electronic glosses, henceforth). Beach et al. (2011) suggest that this new environment provides different ways of approaching a target reading text: it can be hyperlinked

and multimodal (e.g., sounds, pictures, videos, etc.), and can be more interactive than pen-and-paper reading. Along these lines, the research community has explored whether certain formats of electronic textual glossing are more effective than others (see Abraham 2008 for review).

However, studies regarding the relative effects of electronic glosses and their pen-and-paper counterpart (paper glosses, henceforth) are rare. That is, electronic glosses have not yet been shown to be more effective than paper glosses. Laufer (2000) first noted a dearth of related studies, and the issue has still not been sufficiently addressed. In light of this gap, in the present study we compared the effects of electronic and paper glosses on vocabulary learning and reading comprehension in a sample of Korean learners of English as a Foreign Language (EFL) at the university level using an experimental design. We also adopted the cognitive load framework from the psychology literature (Sweller et al. 2011), and used it to evaluate the instructional efficiency of each glossing condition.

Study Background

Theoretical Rationales for Glossing and Literature Review of Traditional Glossing

Glossing can be supported as a supplementary pedagogical device integrated into reading texts by several language acquisition frameworks. As glossed words can be highlighted (Chun 2001) or boldfaced (Roby 1999), glossing can work as a form of input enhancement (Sharwood Smith 1993), increasing the likelihood of target linguistic items being noticed. In light of the noticing hypothesis (Schmidt 2001), it may be suggested that glossing contributes to the acquisition of glossed lexical items in reading texts. Apart from its theoretical implications, glossing is considered relatively easy and does not interrupt reading, compared to looking up words in a dictionary (Nation 2001).

Glossing is thus one of the most commonly recommended practical devices in L2 reading. With respect to vocabulary learning, previous studies (e.g., Hulstijn et al. 1996 on first language (L1) glosses; Jacobs et al. 1994 on L1 and L2 glosses) have reported similar findings on the effects of textual glosses on L2 vocabulary learning, with learners afforded glosses (either L1-based or L2-based) making greater gains than their control counterparts. In these studies, gains in vocabulary knowledge were attributed to glossary information linked to target words. In fact, it is rather difficult to identify any studies reporting that paper glosses are ineffective for L2 vocabulary acquisition.

With respect to improving L2 reading comprehension, however, findings related to the effects of glossing have not been positive (e.g., Jacobs et al. 1994; Johnson 1982). For example, in Jacobs et al. (1994), participants were asked to recall the content of a text after reading. The L1 glossing, L2 glossing, and control conditions yielded similar results. Johnson (1982) exposed advanced undergraduate students of English to different reading conditions, with or without glossary information (i.e., L2 definition). The provision of glossary information did not make any significant differences in terms of reading comprehension, whereas participants' prior knowledge about the text did contribute to comprehension.

A review of the literature on glossing in L2 reading shows that providing glossary information for target words contributes to vocabulary learning, whereas it may not enhance L2 reading comprehension. Having reviewed studies on paper glosses, the next section will discuss another area of work, comparing the effects of electronic and paper textual glosses.

Previous Studies on Electronic and Paper Textual Glosses

Electronic glossing has increasingly gained attention in the field of L2 studies, as computer-assisted reading environments allow a wide range of glossing options. However, an important and fundamental question, which we aim to address in the present study, has been overshadowed: whether electronic glosses are more effective than paper glosses. In other words, researchers have examined the effects of electronic glosses without empirically demonstrating that such technology-aided glossing is more effective than paper glossing (e.g., Lee and Lee 2015; Lomicka 1998). To the best of our knowledge, there have been only three studies (Lyman-Hager et al. 1993; Laufer 2000; Bowles 2004), that explored this question directly.

Lyman-Hager et al. (1993) compared computerized and non-computerized reading conditions among intermediate-level learners of French, with the former enhanced by multimedia annotations and the latter including traditional glosses (i.e., definitions of target words). The results showed that computerized glossing brought about greater vocabulary gains. On the other hand, Bowles (2004) examined the effects of computerized and pen-and-paper glosses among undergraduate beginning-level students of Spanish. Three different groups, including a control group, were exposed to an article from a Spanish newspaper and then took a vocabulary recall and reading comprehension test after the reading task. Both the computer and pen-and-paper glossing groups outperformed the control group in reading comprehension and vocabulary tests, but there was no significant difference between the two glossing groups. Lastly, Laufer (2000) examined the effectiveness of electronic glosses with advanced learners of English in Israel, but offered three different options (e.g., L1 translation, L2 definition, and example sentences) for each target word. The paper gloss group was given an L1 translation of target words on the margin of the passage. As in Bowles (2004), the participants read the passage first and then answered questions on the reading comprehension task, after which they were given the vocabulary recall test without prior notice. Those assigned to the electronic gloss group largely relied on the L1 translation, without opting for other lexical information at their disposal, such as example sentences and L2 definitions. On the other hand, the electronic glosses resulted in greater vocabulary recall than paper glosses on both the immediate and delayed vocabulary tests, although this gap was bridged on the delayed test.

The rationale for the present study is drawn not only from the lack of previous studies, but also from difficulties making direct comparisons between these studies for the following reasons. First, there is variation in participant L2 proficiency, as described above. Given that proficiency level is a major variable affecting the outcomes of any L2 studies, differences in language proficiency can make studies incompatible. Second, the studies differ in terms of whether they gave reading comprehension tasks to the participants in the middle of reading (e.g., Laufer 2000) or after reading (e.g., Lyman-Hager et al. 1993). The differences in task demands also contribute to incompatibility among studies, because this variation has impacts on participant mindsets and strategies used to perform the tasks. Third, these studies provided different types of glossary information (e.g., L1 in the case of Bowles and three different options in Laufer), which makes comparisons difficult.

To assess the effects of electronic and paper L2 glosses in L2 reading, future studies should involve replication, which serves the purpose of checking and confirming the robustness of previous findings (Polio and Gass 1997; Valdman 1993). In particular, conceptual replication (i.e., verifying or refining

theories to move the field forward) is more realistic than exact replication (i.e., closely copying the original study to confirm the previous finding) for L2 studies, among which variation across conditions and methodologies is significant (Chun 2012).

In the present study, we propose a more comprehensive methodology, recruiting intermediate-level L2 learners (the largest group across proficiency levels), investigating reading comprehension and vocabulary acquisition, including target lexical items (TLIs) selected from a wide range of frequency bands (e.g., TLIs fell between the first 5 and 10 k word families, most of which were not familiar to our participants), and providing L2 definitions as glossary information (e.g., intermediate-level L2 learners are not expected to have comprehension problems with L2 glosses; Nation 2009). Moreover, we adopted the framework of cognitive load theory, in order to refine previous efforts to compare effects of different types of glosses, to which we will turn next.

Cognitive Load Theory and Glossing Research

Cognitive load theory is a widely adopted theoretical framework from psychology (Sweller et al. 2011), the aim of which is "to use our knowledge of human cognition to provide instructional design principles" (p. vii). In this framework, a participant's subjective perception of the difficulty of a target learning task (or an item) is measured on a Likert scale and is operationalized as the 'mental effort' spent on the task. Instruction designers can assess participants' cognitive load spent on a task, albeit indirectly, and examine whether a newly designed task or treatment is associated with a lower cognitive load than existing tasks or treatments.

Along with data on mental effort and actual performance on a target task, instruction designers can calculate the instructional efficiency of a target intervention. An important pedagogical implication is that, although treatment groups may have the same scores on a certain test after different treatments, the mental effort invested in each treatment may be different. Paas et al. (2003) suggest that "high-task performance associated with low effort" (p. 67) be referred to as high-instructional efficiency, while "low-task performance with high effort" be referred to as low-instructional efficiency.

Only a small number of glossing studies have adopted a cognitive load framework (e.g., AbuSeileek 2011; Chen and Yen 2013; Yeung 1999). For example, Yeung (1999) administered seven-point Likert scale items regarding participants' perception of difficulty of target vocabulary in printed paper reading contexts, and compared the effect of two different glossing formats on reading comprehension and vocabulary learning. The authors found that learners with a higher proficiency benefited more from an integrated format (glossary information presented right above the target words) in terms of vocabulary learning, more so than from the separated format (glossary information presented separately) in terms of reading comprehension; on the other hand, those with lower proficiency showed that the integrated format enhanced reading comprehension, and the separated format boosted vocabulary learning.

Similarly, Chen and Yen (2013) and AbuSeileek (2011) also compared the effects of different gloss formats, but alluded only to cognitive load theory when explaining their findings and did not administer any measures of mental effort. Indeed, some of their findings, such as the lack of significant differences

between different glossing conditions, could have had greater pedagogical implications if they had administered such a measure. Given the limitations of previous studies, in the present study we more directly adopted this framework and employed a measure of mental effort.

Research Questions

The present study proposes the following three research questions:

(1) What are the effects of different reading conditions (i.e., electronic gloss, paper gloss, no gloss) on EFL learners' reading comprehension?

(2) What are the effects of different reading conditions (i.e., electronic gloss, paper gloss, no gloss) on EFL learners' meaning-recall of target vocabulary?

(3) Are there any significant differences in instructional efficiency among different reading conditions?

Methods

The present study was a part of a larger research project[1] that examined the effects of different textual glossing types on EFL learners' vocabulary learning and reading comprehension.

Participants

The participants were 80 Korean undergraduate students. They were randomly assigned to six classes at the beginning of a mandatory English course. The students in all six classes were taught the same curriculum using the same teaching methods and textbook. The students had intermediate English proficiency, with a mean score on the Test of English for International Communication (TOEIC) of 718. All students were found to be competent computer users.

Reading Text and Target Lexical Items

We extracted a text, 'Education: Fact or Myth' (893 words), from *Cutting Edge Advanced* (Cunningham et al. 2003). After replacing some of the words in the original passage with low-frequency words, we gave a vocabulary test consisting of 50 lexical items drawn from the target text to eight students who had similar profiles to those in the main study. We found that 30 English lexical items out of 50 were unfamiliar to more than 50 % of these students. These 30 lexical items were included in the pre-test of vocabulary.

Experimental Conditions

The experimental design of the present study consisted of three different conditions (i.e., two classes per each condition): electronic gloss, paper gloss, and no gloss. L2 definitions were provided as glossary information. In the electronic glossing condition, participants were asked to read the target reading passage on a computer screen. The glossing device in the present study is called 'tooltip' glossing (see Lee and Lee 2013 for full description). Drawing its name from a function embedded in *Microsoft Office*, a tooltip box appears when the user moves the cursor over the target word and disappears when he or she moves the cursor away. As seen in Fig. 1, tooltip glossing does not block the surrounding context of the glossed word, so the reader can view glossary information and its immediate context simultaneously.

In the paper glossing condition, the participants were given a two-page printed text. This text was further enhanced with marginal glosses, which provided definitions of the glossed words (see Fig. 2). In the control condition, the participants were given the same printed text to read, but without glossary information. In all three conditions, the finalized target lexical items were underlined and written in bold, so that the participants would pay attention to them while reading the text.

Testing and Questionnaire

Two types of language tests were employed to assess vocabulary gains and reading comprehension. First, a meaning-recall typed vocabulary test was developed. We asked participants to provide the meaning of ten target lexical items in either L1 (i.e., Korean) or L2: *toddler*, *be beset with*, *endowments*, *vestige*, *idiot savant*, *nocturnal*, *corrode*, *dismayed*, *attest*, and *plethora*, most of which fell within 5 k and 10 k frequency bands. The reading comprehension test consisted of ten multiple choice items, with the questions assessing whether test-takers accurately understood the information described in the target text or not.

A questionnaire was also developed in order to measure the mental effort spent on comprehending the target text and vocabulary. In light of cognitive load theory (Sweller et al. 2011), seven-point Likert scale items were developed, with 7 indicating “extremely easy” and 1 indicating “extremely difficult.” As noted above, the participants’ self-judged difficulty of the target reading text (i.e., how difficult was it to comprehend the target text?) and target vocabulary (i.e., how difficult was it to comprehend the underlined English words or expressions?) were, respectively, operationalized as the amounts of mental effort required for comprehension of the target reading text and vocabulary. These two questionnaire items were adapted from Yeung (1999). The questionnaire also included an item about the level of participant interest in the target reading text (i.e., how interesting was the target text to read?). The preliminary version of the questionnaire was pilot tested with the initial sample of eight students.

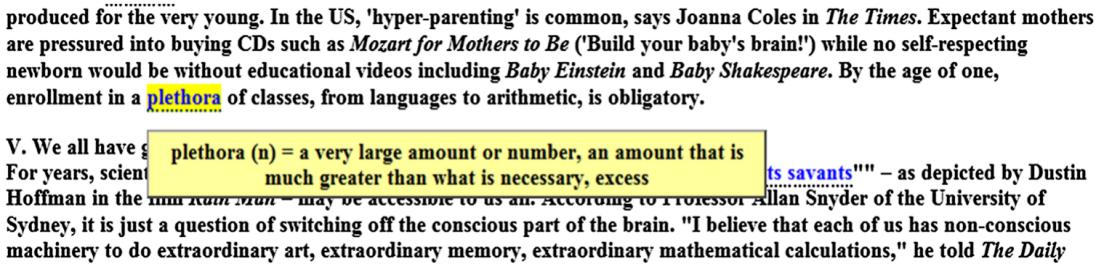


Fig. 1 Screenshot of the electronic gloss

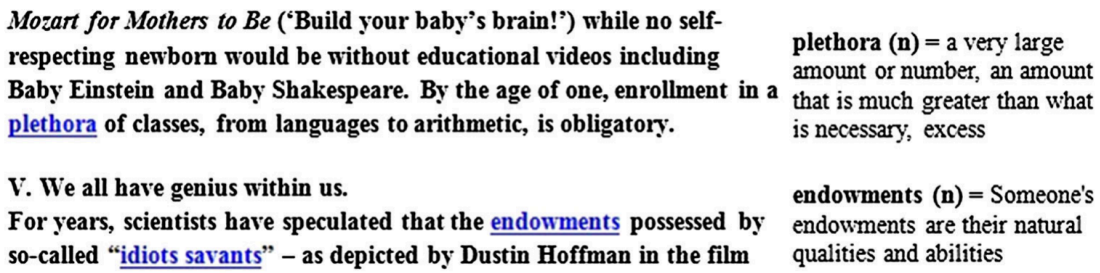


Fig. 2 Illustration of the paper gloss

Procedure

At the outset of the study, the participants were given a vocabulary pre-test, which consisted of 30 English lexical items from the target reading text. Based on the results of this pre-test, ten lexical items that were found to be unfamiliar to more than 90 % of the participants were selected as the final set of target lexical items to be glossed in the present study, because no lexical items were unknown to the entire group of participants.

After 2 weeks, the participants were given the reading comprehension activity. The participants read the text according to their respective reading conditions for 25 min and then completed the reading comprehension test. During this reading comprehension activity, electronic and paper glossing groups were encouraged to use the glossary information if difficult or unknown target lexical items cropped up in their reading. On the other hand, the control group was asked to guess the meaning of the underlined target lexical items by consulting the immediate context. After 25 min, all materials were collected, and web pages were closed. Subsequently, the vocabulary post-test was administered for 3 min without prior notification. Afterward, the participants completed the questionnaire. Two weeks after the reading activity, without prior notice, the delayed vocabulary test was administered to the participants.

Scoring and Data Analyses

Responses on the reading comprehension test, which consisted of ten multiple choice questions, were marked right (1 point) or wrong (0 point), with the maximum score being 10 points. For the vocabulary test, two points were given for a completely correct meaning of the target word, while one point was given for a partially correct meaning (i.e., semantically approximate answer). Two raters marked the vocabulary test, with 20 % of the testing sheets scored by both raters to evaluate inter-rater reliability. The reliability was found to be 0.94 (Cohen's Kappa, $p < 0.05$).

In order to make assumptions about the parameters of the population distribution, we used Levene's and Shapiro–Wilk tests to confirm the homogeneity of variances as well as distributions of scores of the three groups on the reading comprehension and vocabulary tests. Except for the normality of the paper glossing group on the post-test of vocabulary, the results of these tests were not significant, indicating that our dataset satisfied the requirements of parametric tests.

For the first research question regarding reading comprehension, a simple regression analysis was implemented to compare the three groups' performance on the reading comprehension test, and their perceptions of the difficulty of, and level of interest in, the target text.

As for the second research question on vocabulary learning, a multiple regression analysis was used, with (1) the experimental condition, (2) vocabulary pre-test, and (3) English proficiency (i.e., TOEIC scores) included as independent variables. Given that the post-test variable may be non-normal (due to the paper glossing group), this regression involved the Huber-White estimator to control possible heteroscedasticity.

For the third research question regarding the instructional efficiency of each condition, the participants' responses to seven-point Likert scale items were reversed in light of Yeung (1999), with one indicating 'extremely easy' and seven referring to 'extremely difficult.' Then, new variables were created using the 'instructional efficiency' formula from Paas and van Merriënboer (1993). The instructional efficiencies of the three reading conditions on the post-test and delayed test were compared using one-way ANOVA.

Due to our small sample size, we set the significance levels at 10, 5, and 1 %.

Results

Performance on a Reading Comprehension Test and Perception of the Target Reading Text

The first research question concerns reading. Table 1 presents results for group performance on the reading comprehension test as well as perceptions toward the target reading text. In this section, EG, PG, and CG refer to electronic glossing, paper glossing, and control group, respectively.

Simple regression analysis showed that the differences among the three groups for reading comprehension test scores did not reach statistical significance ($b = 0.44$, $p > 0.10$). Likewise, there were no significant differences among the three groups in levels of interest in the target text ($b = -0.15$, $p > 0.10$) or perceived difficulty of the target text ($b = -0.14$, $p > 0.10$). These findings indicate that the differences in reading conditions did not lead to any significant differences in reading when the density rate of unfamiliar words in the target text was low (approximately 5 %).

Performance on Vocabulary Tests

The second research question concerns vocabulary development through different reading conditions. Descriptive statistics for group performance at three testing points are presented in Table 2, along with the results of paired *t* tests comparing group performance on these tests.

A series of paired *t* tests showed a consistent pattern among the three groups regarding performance at three testing points. All groups made significant gains on the post-test, which was taken immediately after the reading activity, and these gains were retained for another 2 weeks.

A multiple regression analysis with Huber-White estimators was performed to statistically compare the three groups' performances on the post-test and delayed test (see Table 3). In the case of the post-test, there was a significant effect of reading conditions on performance ($b = -1.58$, $p < 0.001$; see Model 1 in Table 3), after controlling for the effects of the pre-test and of English proficiency. When including dummy variables for different conditions (i.e., EG, PG, CG), the mean score of EG on the post-test was significantly higher than that of PG ($b = 1.60$, $p < 0.10$; see Model 2 in Table 3). The mean score of PG was significantly higher than that of CG ($b = -1.55$, $p < 0.10$; see Model 2 in Table 3).

Table 1 Descriptive statistics for the reading comprehension test and perceptions of the reading text

Group	Reading comprehension test	Level of interest in the target text	Perception of the difficulty of the target text
	Mean (SD)	Mean (SD)	Mean (SD)
EG ($n = 29$)*	5.28 (1.70)	4.41 (1.28)	4.74 (1.10)
PG ($n = 27$)	5.04 (2.33)	4.70 (1.20)	4.81 (1.04)
CG ($n = 24$)	6.21 (1.67)	4.08 (1.59)	4.46 (1.10)

* As two participants failed to submit their questionnaires online due to technical issues, the number of the participants for questionnaire items in Tables 1 and 4 was 27

Table 2 Descriptive statistics for vocabulary tests and results of *t* tests comparing performance using three testing periods

Group	Pre-test (max = 20.0) Mean (SD)	Post-test (max = 20.0) Mean (SD)	Delayed test (max = 20.0) Mean (SD)	*t* tests	*t* value
EG (*n* = 29)	0.83 (0.85)	7.24 (3.24)	7.03 (4.06)	Pre < post	9.98*
				Pre < delayed	7.84*
				Delayed ≒ post	0.27
PG (*n* = 27)	1.11 (1.34)	5.96 (3.74)	5.33 (3.26)	Pre < post	8.24*
				Pre < delayed	7.46*
				Delayed ≒ post	0.97
CG (*n* = 24)	1.13 (0.95)	4.42 (2.87)	4.25 (3.21)	Pre < post	5.98*
				Pre < delayed	5.04*
				Delayed ≒ post	0.42

* $p < 0.006$ (Bonferroni correction applied)

Table 3 Coefficient values of regression models for vocabulary post-tests and delayed tests

Independent variables	Dependent variable: post-test		Dependent variable: delayed test	
	Model 1	Model 2	Model 3	Model 4
Group	−1.577*** (0.432)		−1.528** (0.507)	
EG		1.599† (0.860)		1.997* (0.954)
CG		−1.554† (0.798)		−1.034 (0.839)
Pre-test	1.083** (0.355)	1.083** (0.357)	0.621† (0.359)	0.632† (0.358)
English proficiency	−0.001 (0.004)	−0.001 (0.004)	−0.005 (0.005)	−0.005 (0.005)
Constant	8.296** (2.983)	5.135† (2.938)	11.192** (3.549)	7.986* (3.487)
R^2	0.219	0.219	0.142	0.146

Standard errors are in parentheses. *PG* is the omitted condition category in Models 2 and 4

† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

In the case of the delayed test, there was also a significant effect of reading conditions on performance ($b = -1.53$, $p < 0.01$; see Model 3 in Table 3), after controlling for the effects of the pre-test and of English proficiency. The results were similar to those of the post-test: the mean score of EG was significantly higher than that of PG ($b = 2.00$, $p < 0.05$; see Model 4 in Table 3). On the other hand, the mean score of PG was not different from that of CG ($b = -1.03$, $p > 0.10$; see Model 4 in Table 3), indicating that the effect of PG faded away after the experiment.

Instructional Efficiency of the Three Reading Conditions for Learning Target Vocabulary

Finally, the level of mental effort (i.e., cognitive load) spent on comprehending the target vocabulary as well as instructional efficiencies for each reading condition on the post-test and delayed test are presented in Table 4.

The three groups were found to experience similar amounts of cognitive load dealing with the target vocabulary, finding the words to be slightly challenging. In terms of instructional efficiency, EG resulted in greater efficiency than the other two conditions on both post-test and delayed tests, indicating that it was the most efficient of the three conditions for enhancing vocabulary learning. The results of one-way ANOVA showed that there were significant effects of reading conditions in terms of instructional efficiency, both for the post-test $F(2, 75) = 4.80$, $p < 0.05$ and for the delayed test, $F(2, 75) = 4.12$, $p < 0.05$. These findings were found to be driven primarily by differences between EG and CG, which showed the largest differences in instructional efficiency.

Table 4 Descriptive statistics for mental effort regarding target vocabulary and instructional efficiency for post-tests and delayed tests

Group	Mental effort on target vocabulary Mean (SD)	Instructional efficiency (post-test) Mean (SD)	Instructional efficiency (delayed test) Mean (SD)
EG (*n* = 27)	5.07 (0.99)	0.38 (0.93)	0.39 (1.11)
PG (*n* = 27)	5.30 (0.82)	−0.03 (1.19)	−0.09 (1.04)
CG (*n* = 24)	5.42 (0.97)	−0.44 (1.06)	−0.38 (1.10)

Discussion and Conclusion

The three different reading conditions did not result in significant differences in reading comprehension, a result that corroborates the findings of Jacobs et al. (1994) and Johnson (1982). A possible explanation for this finding was proposed by Hu and Nation (2000), who suggested that a reader may gain an adequate level of comprehension when they have approximately 98 % text coverage (i.e., the proportion of the running words they already know prior to reading). Based on the results of piloting and the vocabulary pre-test, we found that the proportion of unfamiliar vocabulary in the target text was generally low, approximately 5 % at most, suggesting that no group in the present study was disadvantaged in terms of comprehending the target text (Nation 2001). Furthermore, two glossing conditions or lack of glossing did not lead to significant differences in the participants' levels of interest in, or perceptions of difficulty of, the target text.

With regard to vocabulary learning, the participants in the electronic glossing condition group fared better than the paper-based glossing group as well as the control group, and this difference was maintained 2 weeks after the reading task. However, the superior performance that the participants in the paper-based glossing condition showed at the post-test faded away at the delayed-test time point. Our findings related to cognitive load lend weight to this argument, as the instructional efficiency of the electronic glossing condition was the greatest.

Based on our findings, we suggest that the participants in the present study, who were reportedly competent users of computers and intermediate-level L2 learners, adapted well to computer-assisted reading environments and benefited from the electronic glosses therein. Perhaps this finding may not be generalizable to learners who have difficulty processing online texts (Kress 2003), which is inherently different from reading hard copy text in several ways. While more evidence on the positive effects of electronic textual glosses is needed, having learners with intermediate or higher proficiency levels in the target language and adequate levels of computer literacy may be seen as a green light for using electronic glosses, in light of our findings and those of Laufer (2000) and Lyman-Hager et al. (1993).

Some limitations of the present study are worth mentioning. First, we only examined one aspect of vocabulary knowledge—meaning recall. It is possible that the three reading conditions examined in this study may bring about differences in learning other aspects of vocabulary knowledge (e.g., form), and this question should be explored in future research. Second, cognitive load was measured using self-reported questionnaire items, which may not precisely measure actual cognitive load. Future studies may seek other methods of measuring mental effort spent on the target task, which would lead to a more accurate figure of instructional efficiency regarding L2 glossing.

Our results have important implications for future research. As the result of a conceptual replication approach, the findings of this study lend support to the effectiveness of electronic glossing (compared to paper-based glossing). In this way, it consolidates the theoretical and practical foundations of previous efforts, and suggests future directions to introduce and adopt various types of glosses in computer-based reading environments. Moreover, unlike most previous studies on glossing, in this study, we adapted direct cognitive load measures and were able to take a step forward from simply suggesting that electronic glossing is effective: we showed that it is efficient as well. Indeed, most research in this field has predominantly focused on outcomes (i.e., the amount of learning), while ignoring how much effort each learner spends on a target task. However, even if the learning outcomes of two groups are the same, this does not necessarily imply that the groups went through the same learning process. Cognitive load theory offers a point of view from which we can examine what learners experience during a given learning task. We also expect that the generation of instructional efficiency figures based on this framework will allow us to compare the effectiveness of different glossing conditions across a wide range of studies in a more direct manner.

Examining instructional efficiency is useful for learners as well as material and instruction designers. That is, given a wide variety of multimodal options (e.g., L1 equivalents, L2 definition, example sentences, pictures, sounds, videos, etc.) available in computerized reading (Beach et al. 2011), material designers may wonder which types of glossary information will be more effective and efficient than others. With more evidence to come, instructional efficiency can be employed to suggest the most effective and efficient types of glossary information and formats.

For future studies, the use of qualitative research instruments such as think-aloud protocols or stimulated recall may enable us to explore additional potential benefits of electronic glosses and computer-assisted reading environments. We believe that this will be a useful starting point for understanding what truly differentiates computer-assisted reading from its traditional counterpart. Another line of research that is worth considering is the effects of electronic glosses and computer-assisted environments on learners with different levels of computer literacy and different amounts of experience with electronic glossing and computer-assisted reading. This is another important topic that may provide valuable pedagogical implications for a wide range of EFL and ESL classrooms with different learner populations.

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1. The data related to the electronic glossing group's performance on the vocabulary test and questionnaire items reported in the present study were presented elsewhere in Lee and Lee (2015).↩